

Deliverables

Spring 2025-2026

Power-point presentation

Object Detection / Instance Segmentation

الاسم	model
1	سما أحمد
2	اسراء نصر
3	شهد ايمن
4	مازن
5	أحمد أيمن
6	محمود خالد
7	بلال
8	هيا
9	سما محمد سليمه
10	محمد ابراهيم

Power-point presentation - Structure

	Section	Suggested Discussion Points
1	Introduction	• Why the selected algorithm was proposed • Problems in earlier methods that motivated this model
2	Historical Background	• Previous object detection • Position in Literature
3	Main Contribution	• Core innovation of the model • What makes it different from earlier detectors • Novel modules or design choices
4	Model Architecture, with Diagram	• Backbone network • Neck structure • Detection head
5	How the Model Works	• Input preprocessing, Feature extraction • Bounding box prediction • Non-Maximum Suppression (NMS), etc.
6	Output Representation & Visual Illustration	• Bounding boxes • Class labels • Confidence scores • Detection examples (inference mode) • Visualization of correct vs. incorrect detections

Power-point presentation - Structure

	Section	Suggested Discussion Points
7	Benchmark Datasets	• COCO • Pascal VOC • Open Images • VisDrone • CrowdHuman .. etc.
8	Strengths	• Detection accuracy • Real-time capability • Robustness • Scalability • Small-object handling etc.
9	Comparison with Other Algorithms	• Accuracy-speed tradeoff • Complexity comparison (mAP (mean Average Precision), FPS/inference speed, IoU threshold), etc.
10	Limitations	• Occlusion problems • Small object detection issues • Computational cost • Sensitivity to lighting/background , etc.
11	Real world applications	• Autonomous driving • Surveillance • Medical imaging • Robotics • Retail analytics • Smart cities , etc.
12	Future research directions	• Open problems • Efficiency improvements • Edge deployment • Multi-modal detection
13	Your review	You have to present your review and critic for the algorithm

Research Paper

There will be a discussion for each project that will check all the required deliverables .

Object detection / instance segmentation

الاسم	Team	Datasets Category
1	سما أحمد	
2	اسراء نصر	
3	شهد ايمن	
4	مازن	
5	أحمد أيمن	
6	محمود خالد	
7	بلال	
8	هيا	
9	سما محمد سليمه	
10	محمد ابراهيم	

Requirements

- **Team Formation:** team should consist of **two members**.
- **Selection of Application Category:** team must select **one category** such as: aerial imagery, drone-based imagery, autonomous driving, medical imaging, indoor object detection, surveillance, agriculture, etc. Each has 2 datasets.
- **Literature Review:** Conduct a review **of recent object detection methods related** to the selected application category and datasets. The review should explain current challenges and research trends.
- **Selection of Object Detection Models:** The team must state, use, and evaluate **four object detection algorithms**. One of the selected models must be a **YOLO-based model** (e.g., YOLOv10, YOLOv11, or newer versions), while the others, would be algorithms relevant to **2-stages** algorithm, **transformer-based**, the last one **could be based on hybrid-approach**.

Requirements

- **Environment Setup:** Prepare the development and training environment including frameworks, GPU setup, libraries, and dataset preprocessing pipelines.
- **Dataset Preprocessing:** Analyze dataset characteristics including number of classes, object distribution, annotation formats, image resolutions, and potential challenges such as occlusion or small objects.
- **Model Fine-Tuning:** Fine-tune the four selected object detection models using the two chosen datasets. Training configurations and hyper-parameters should be clearly documented.
- **Inference and Object Detection:** Use the trained models to detect and classify objects from the selected datasets. Provide qualitative visual results using detection examples.
- **Performance Evaluation:** Evaluate all models using appropriate metrics such as: mAP, Precision, Recall, F1-score, IoU, inference speed (FPS), model size, and computational complexity.

Requirements

- **Comparative Study:** Perform a detailed comparative analysis between the four models across both datasets. Discuss strengths, weaknesses, accuracy-speed tradeoffs, and suitability for the selected application domain.
- **Result Visualization:** Present results using tables, graphs, confusion matrices, loss curves, and sample detection outputs.
- **Research Paper Preparation:** State, write a conference- or journal-style paper that includes: introduction, related work, methodology, datasets, experiments, results, discussion, limitations, and conclusions.
- **Discussion of Application Importance:** The paper should clearly explain the importance of object detection techniques within the selected application domain and justify the choice of datasets and models.

Requirements

- **GitHub Repository Development:** Create a GitHub repository containing: source code, trained models, dataset preparation instructions, experiment configurations, comparative results, and relevant documentation.
- **Repository Documentation:** The repository must include a professional README file explaining installation steps, training procedures, inference instructions, evaluation methods, and project structure.
- **Reproducibility Requirement:** All experiments should be reproducible. The repository should include configuration files, dependency lists, and clear execution commands.
- **Final Deliverables:** Final submission must include:
 - 1) Research paper
 - 2) GitHub repository materials & share link with course instructor, it must have:
 - i. Models (fine-tuned)
 - ii. Dataset sample
 - iii. Comparative results

Surveys papers and Comparative Studies

A **Comparative Study** of Various Object Detection Algorithms and Performance Analysis

- <https://ijcseonline.org/index.php/j/article/view/6195> (**2020**), **Cited by 55**
- “A comparison of these algorithms is done and also their results as well as performance is analysed. The performance and exactness should be of utmost important in analysing the algorithms. Object detection can be done through various techniques like **R-CNN, Fast R-CNN, Faster R-CNN, Single Shot detector (SSD)** and **YOLO v3**. A comparison of these algorithms is done”

Comparative analysis of deep learning image detection algorithms

- <https://link.springer.com/article/10.1186/s40537-021-00434-w> (**2021**), **Cited by 499**
- “The results obtained by comparing the effectiveness of these algorithms (**SSD, Faster-RCNN, and YOLO**) on the same dataset (Microsoft COCO dataset) can help gain an insight on the unique attributes of each algorithm, understand how they differ from one another and determine which method of object recognition is most effective for any given scenario”

A **survey** of modern deep learning based object detection models

- <https://www.sciencedirect.com/science/article/abs/pii/S1051200422001312> (**2022**), **Cited by 1388**
- “It also covers contemporary lightweight classification models used **on edge devices**. Lastly, we compare the performances of these architectures on multiple metrics. We compare the performance of various object detectors on PASCAL VOC 2012 [36] and Microsoft COCO [22] datasets. Performance of object detectors is influenced by a number of factors like input image size and scale, feature extractor, GPU architecture, number of proposals, training methodology, loss function, etc., which makes it difficult to compare various models without a common benchmark environment”

A **survey** of modern deep learning based object detection models

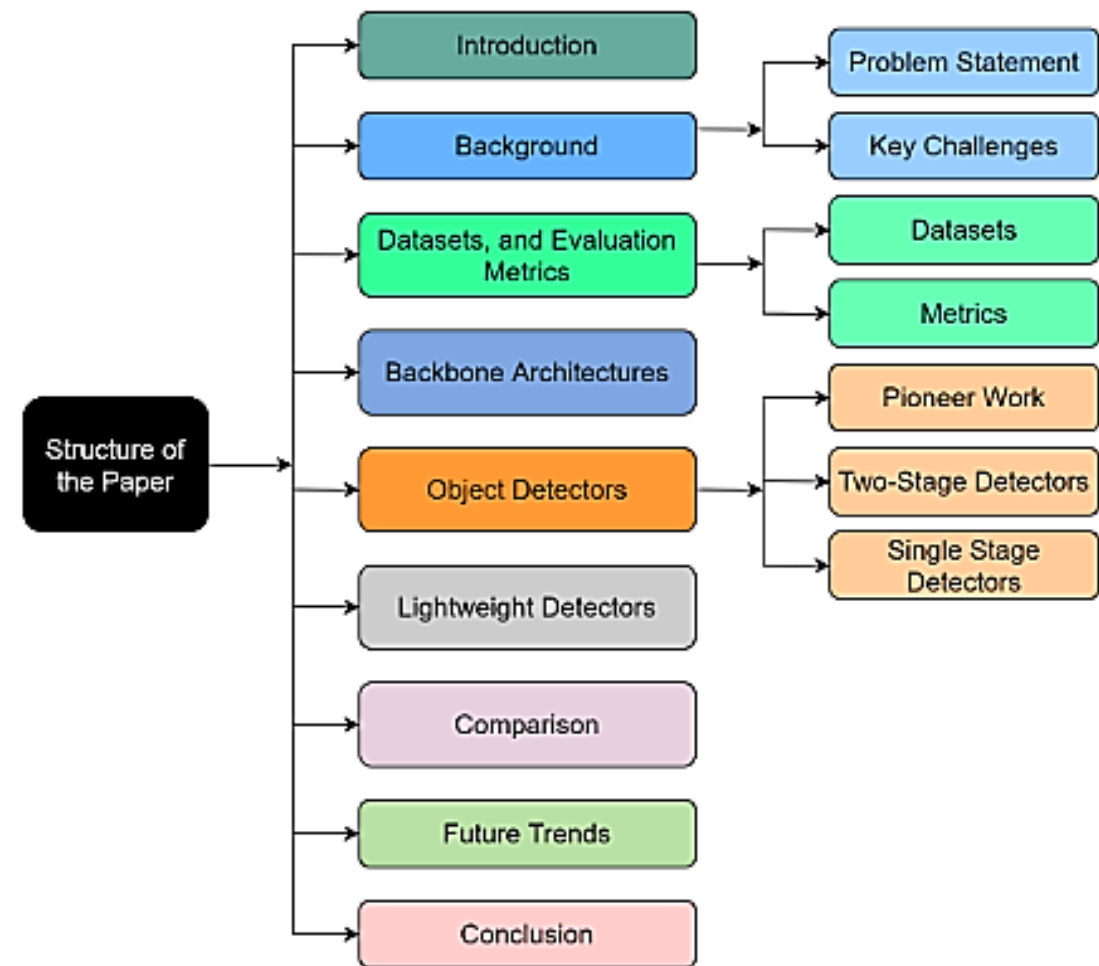


Fig. 1: Structure of the paper.

TABLE II: Comparison of Backbone architectures.

Model	Year	Layers	Parameters (Million)	Top-1 acc%	FLOPs (Billion)
AlexNet	2012	7	62.4	63.3	1.5
VGG-16	2014	16	138.4	73	15.5
GoogLeNet	2014	22	6.7	-	1.6
ResNet-50	2015	50	25.6	76	3.8
ResNeXt-50	2016	50	25	77.8	4.2
CSPResNeXt-50	2019	59	20.5	78.2	7.9
EfficientNet-B4	2019	160	19	83	4.2

A **survey** of modern deep learning based object detection models

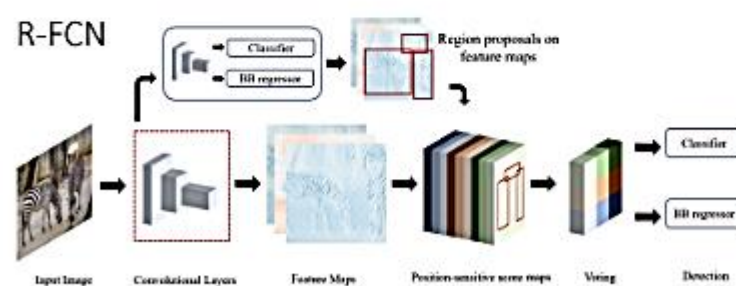
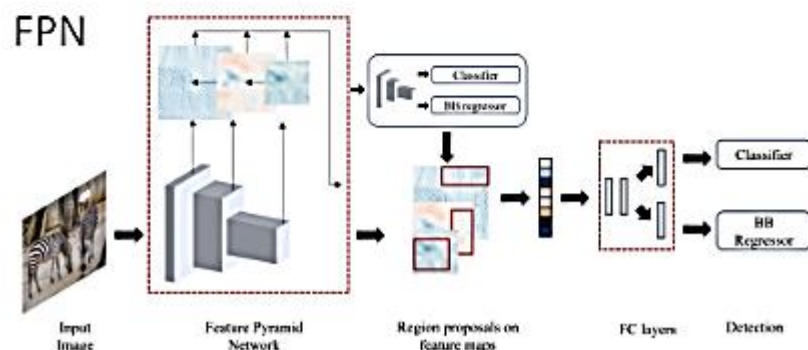
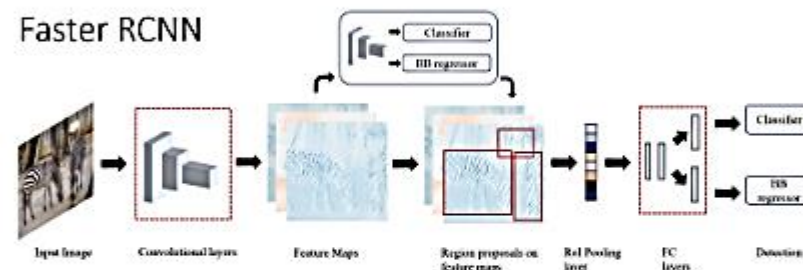
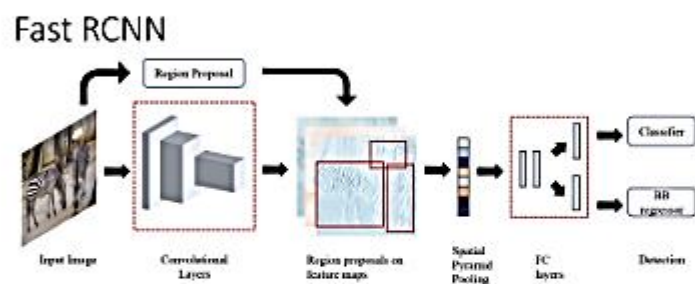
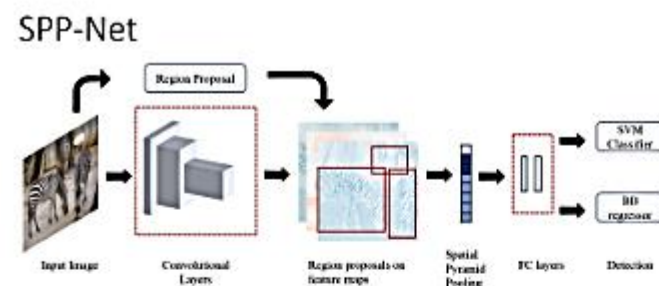
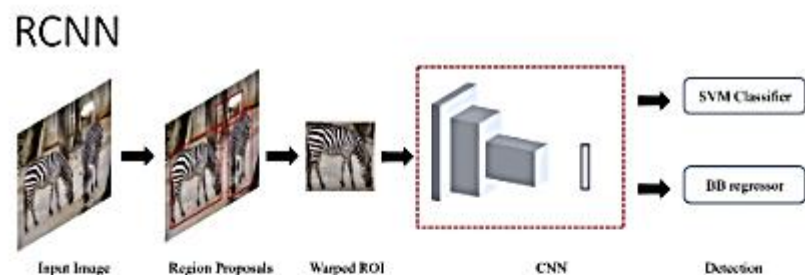


Fig. 8: Illustration of the internal architecture of different two stage object detectors^[2]

A **survey** of modern deep learning based object detection models

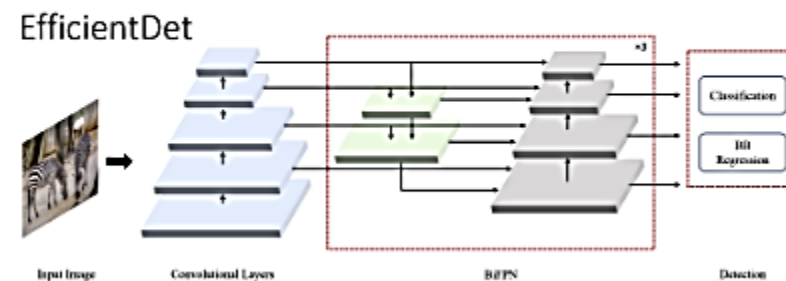
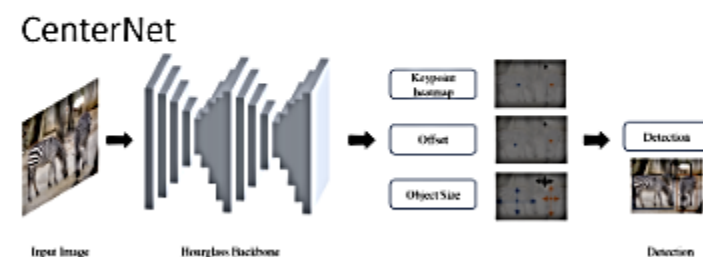
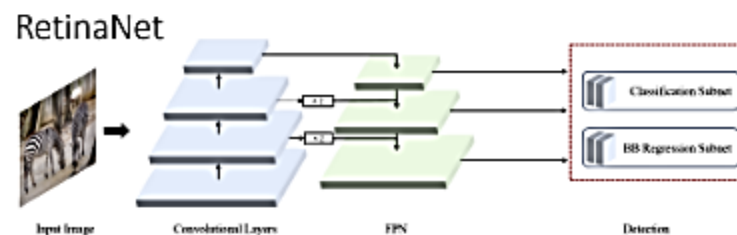
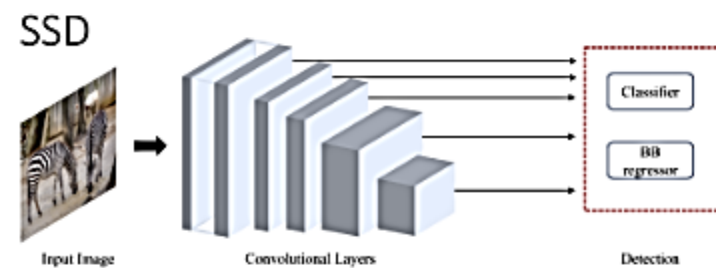
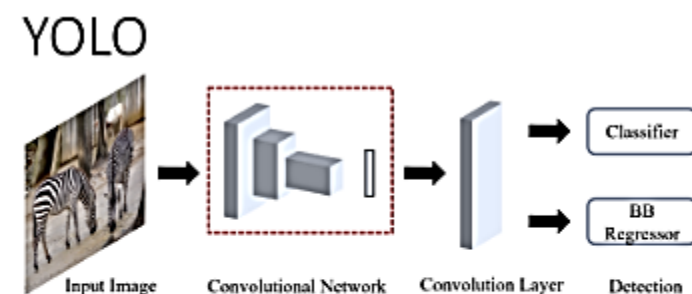
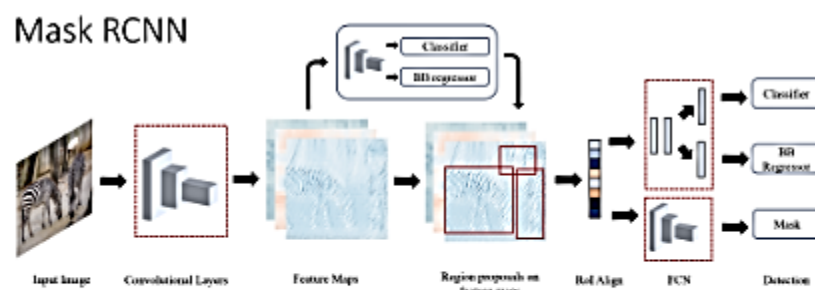


Fig. 9: Illustration of the internal architecture of different two and single stage object detectors².

A **survey** of modern deep learning based object detection models

Model	Year	Backbone	Size	AP _[0.5:0.95]	AP _{0.5}	FPS
R-CNN*	2014	AlexNet	224	-	58.50%	~0.02
SPP-Net*	2015	ZF-5	Variable	-	59.20%	~0.23
Fast R-CNN*	2015	VGG-16	Variable	-	65.70%	~0.43
Faster R-CNN*	2016	VGG-16	600	-	67.00%	5
R-FCN	2016	ResNet-101	600	31.50%	53.20%	~3
FPN	2017	ResNet-101	800	36.20%	59.10%	5
Mask R-CNN	2018	ResNeXt-101-FPN	800	39.80%	62.30%	5
DetectoRS	2020	ResNeXt-101	1333	53.30%	71.60%	~4
YOLO*	2015	(Modified) GoogLeNet	448	-	57.90%	45
SSD	2016	VGG-16	300	23.20%	41.20%	46
YOLOv2	2016	DarkNet-19	352	21.60%	44.00%	81
RetinaNet	2018	ResNet-101-FPN	400	31.90%	49.50%	12
YOLOv3	2018	DarkNet-53	320	28.20%	51.50%	45
CenterNet	2019	Hourglass-104	512	42.10%	61.10%	7.8
EfficientDet-D2	2020	Efficient-B2	768	43.00%	62.30%	41.7
YOLOv4	2020	CSPDarkNet-53	512	43.00%	64.90%	31
Swin-L	2021	HTC++	-	57.70%	-	-

*Models marked with * are compared on PASCAL VOC 2012, while others on MS COCO. Rows colored gray are real-time detectors (>30 FPS).

Comparative Study of Some Deep Learning Object Detection Algorithms: R-CNN, FAST R-CNN, FASTER R-CNN, SSD, and YOLO

- <https://www.bibliomed.org/mnsfulltext/262/262-1681947210.pdf?1778517722> (**2023**), **Cited by 38**
- “Findings from the performed analysis indicate that **YOLO** stands out as the predominant algorithm for drone surveillance among different deep learning models used in various application fields and being a one-stage detector. In terms of usage in **traffic-related applications**, **SSD** proved to be a prominent one-stage detector alongside **Faster R-CNN** which gained popularity as a two-stage detector preferred for applications in **the medical field**”

A **Comparative Analysis** of Modern Object Detection Algorithms: YOLO vs. SSD vs. Faster R-CNN

- <https://syekhnurjati.ac.id/journal/index.php/itej/article/view/123> (**2023**),
Cited by 66
- “Each algorithm is tested under similar conditions to ensure a fair comparison. The results indicate that while **YOLO** excels in real-time applications due to its high speed, **SSD** offers a middle ground with respectable accuracy and speed, making it suitable for applications requiring a balance of both. **Faster R-CNN** demonstrates superior accuracy, making it ideal for scenarios where detection precision is paramount, despite its slower performance. This comparative analysis highlights the strengths and weaknesses of each algorithm, providing valuable insights for researchers and practitioners in selecting the appropriate object detection method for their specific needs.”

Performance analysis of deep learning-based object detection algorithms on COCO benchmark: a **comparative study**

- <https://link.springer.com/article/10.1186/s44147-024-00411-z> (**2024**), **Cited by 69**
- “Dynamic head (DyHead) stands out as the superior algorithm, particularly at medium IoU thresholds, making it suitable for precise object detection in **smart city applications**. The performance analysis using Average Precision at Medium IoU is reinforced by the Average Precision at Low IoU (APL), consistently depicting DyHead’s superiority. These findings provide valuable insights for researchers and practitioners, guiding them toward employing DyHead for tasks prioritizing accurate object localization and classification in smart cities. Moreover, we analyze by illustrating the mAP results of object detection algorithms from 2016 onwards, based on **Multi-scale, Single-scale, ResNet, FPN, DCN**, and **YOLO** networks.”

2D Object Detection: A **Survey**

- <https://www.mdpi.com/2227-7390/13/6/893> (**2025**), **Cited by 11**
- “The review systematically categorizes contemporary object detection methods into three key paradigms: one-stage, two-stage, and transformer-based, highlighting their development milestones and core contributions. The paper provides an in-depth analysis of each paradigm, detailing landmark methods and their impact on the progression of the field”

A **Review** of Object Detection Techniques in IoT-Based Intelligent Transportation Systems

- <https://www.sciencedirect.com/org/science/article/pii/S1546221825005211> (**2025**),
Cited by 6
- “we adopted the Preferred Reporting Items for **Systematic Reviews and Meta-Analyses (PRISMA)** approach to search, screen, and assess the eligibility of relevant literature, ultimately including 88 studies. Through an analysis of these studies, we summarized the characteristics, advantages, and limitations of object detection technology across the traditional methods stage and the deep learning-based methods stage. Additionally, we examined its applications in ITS from three perspectives: vehicle detection, pedestrian detection, and traffic sign detection. We also identified the major challenges currently faced by these technologies and proposed future directions for addressing these issues. This review offers researchers a comprehensive perspective, identifying potential improvement directions for object detection technology in ITS, including accuracy, robustness, real-time performance, data annotation cost, and data privacy.”

Object Detection Algorithms - Taxonomy

CNN based-architectures

Single stage

- YOLO (You Only Look Once)
- SSD (Single Shot Detection)
- RetinaNet
- EfficientDet (Efficient detection)
- CenterNet
- CornerNet
- FCOS — Fully Convolutional One-Stage Object Detection

Two stages

- R-CNN (Regions with Convolutional Neural Networks)
- Fast R-CNN
- Faster R-CNN

Transformer-based Architectures

- DETR (DEtection TRansformer)
- Deformable DETR
- DINO (DETR with Improved DeNoising Anchor Boxes)
- Swin Transformer
- ViTDet (Vision Transformer for Detection)

Transformer extension based-architectures

Hybrid CNN with transformer

- COATNet (Convolution and Attention Network)
- TransYOLO (hybrid model that combines the YOLO (You Only Look Once) real-time detection framework with Transformer)
- TSP-FCOS (Transformer-based Set Prediction with FCOS)(FCOS is Fully Convolutional One-Stage Object Detection)

Attention-based detectors

- CBAM-based (Convolutional Block Attention Module-based)
- Attention R-CNN
- Dual attention Networks
- SE-enhanced detectors (Squeeze-and-Excitation (SE) blocks)

Multi-scale object detection-based Architectures

- BiFPN segmentation (Bi-directional Feature Pyramid Network)
- NAS-FPN (Neural Architecture Search for Feature Pyramid Network)

Instance Segmentation Taxonomy

CNN-based Architectures

Single stage

- YOLACT (You Only Look At CoefficientTs)
- YOLACT++
- SOLO (Segmenting Objects by LOcations)
- SOLOv2
- CondInst (Conditional Convolutions for Instance Segmentation)
- U-Net
- 3D U-Net

Two stages

- Mask R-CNN (Mask Region-based Convolutional Neural Network)
- Cascade Mask R-CNN
- PANet (Path Aggregation Network)
- PointRend (Point-based Rendering)

Transformer-based Architectures

- Mask2former (Masked-attention Mask Transformer)
- Maskformer (mask classification-based image segmentation framework)
- Swin transformer (Shifted Window-Based Self-Attention)
- Segmenter (specific transformer-based model architecture used for image segmentation)

Transformer-extension based Architectures

Hybrid

- Swin Mask R-CNN
- TransUNet (Transformer-based U-Net)
- Segformer (Segmentation Transformer)

Attention-based

- Dual attention Network
- Attention UNet

Multi-scale feature segmentation -based Architectures

- BiFPN segmentation (Bi-directional Feature Pyramid Network.)
- HRNet segmentation (High-Resolution Network segmentation)

Datasets Categories

Datasets Sources

<https://docs.ultralytics.com/datasets>

<https://datasetsearch.research.google.com/>

<https://huggingface.co/datasets>

<https://www.kaggle.com/datasets>

<https://data.gov/>

<https://data.pnnl.gov/>

Object Detection: Aerial images

xView Dataset

<https://docs.ultralytics.com/datasets/detect/xview>



DOTA Dataset with OBB

<https://docs.ultralytics.com/datasets/obb/dota-v2>



Object Detection: Drone-based image and video analysis

VisDrone Dataset

<https://docs.ultralytics.com/datasets/detect/visdrone>



Wider Person

<https://www.kaggle.com/datasets/imneonizer/wider-person>



Object Detection: Fire datasets

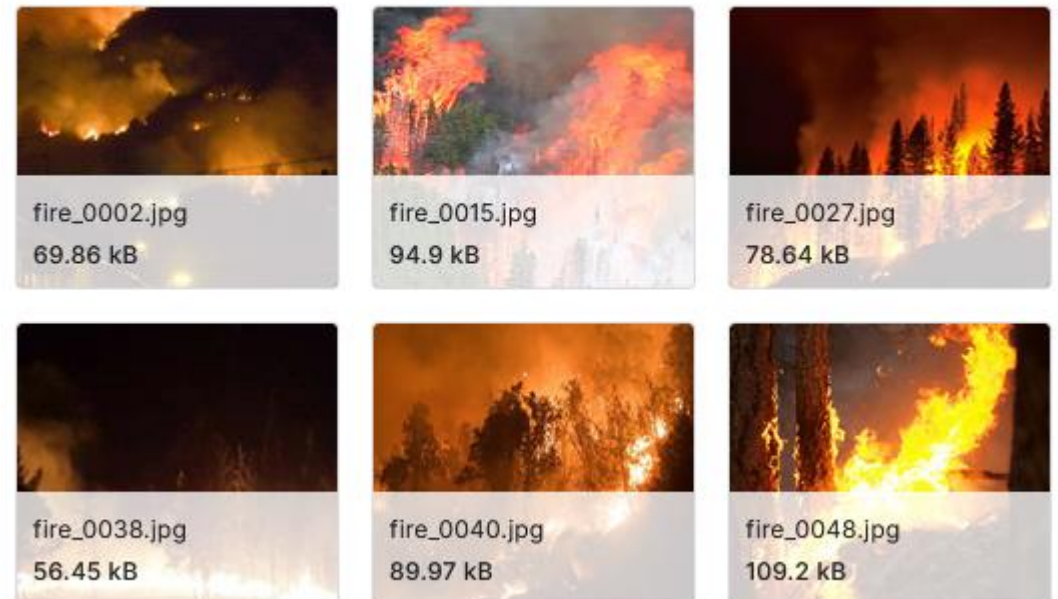
Forest Fire Dataset

<https://etsin.fairdata.fi/dataset/1dce1023-493a-4d63-a906-f2a44f831898>



Forest Fire Dataset

<https://www.kaggle.com/datasets/alik05/forest-fire-dataset>



Object Detection: Autonomous driving

xView Dataset

(<https://www.ultralytics.com/glossary/autonomous-vehicles>):

<https://docs.ultralytics.com/datasets/detect/kitti>

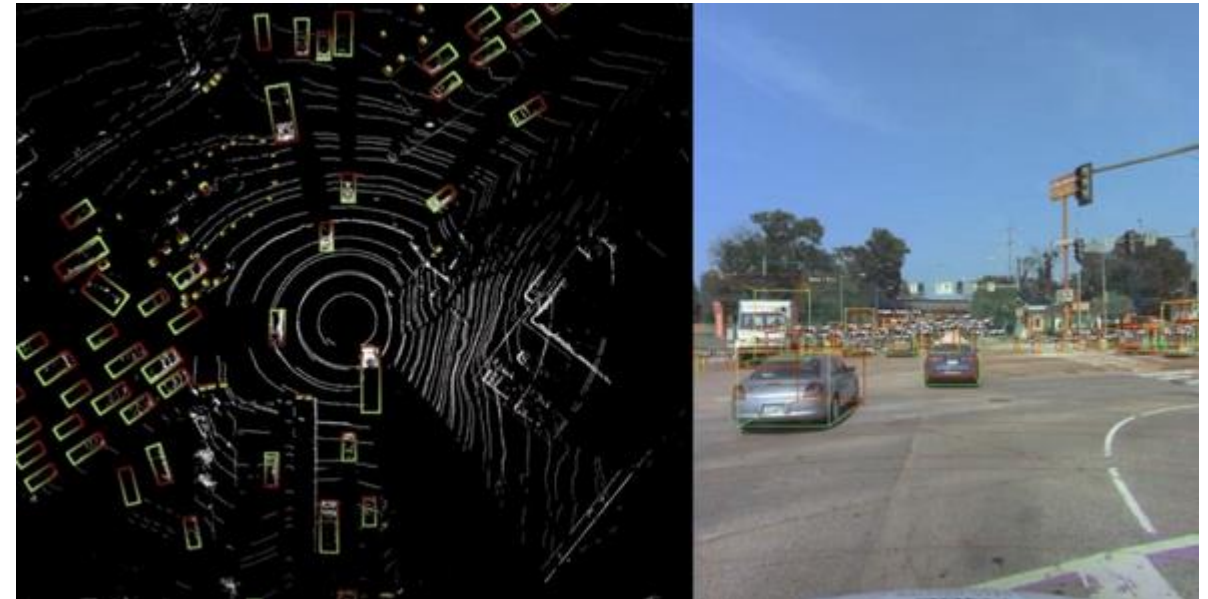


DOTA Dataset with OBB

<https://docs.ultralytics.com/datasets/detect/argoverse>

<https://www.argoverse.org/av2.html>

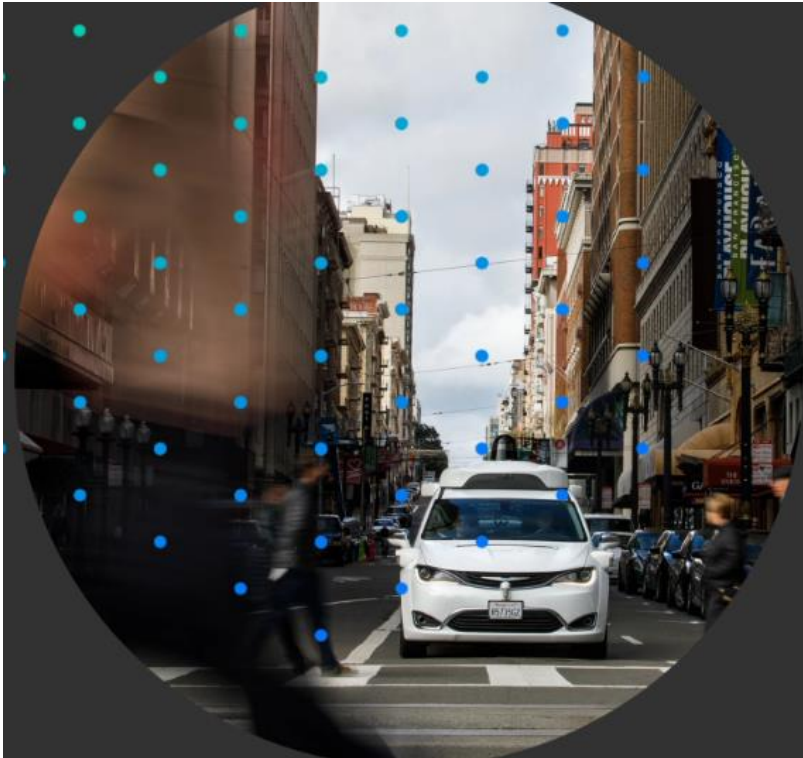
<https://docs.ultralytics.com/datasets/detect/argoverse>



Object Detection: Autonomous driving

waymo Dataset

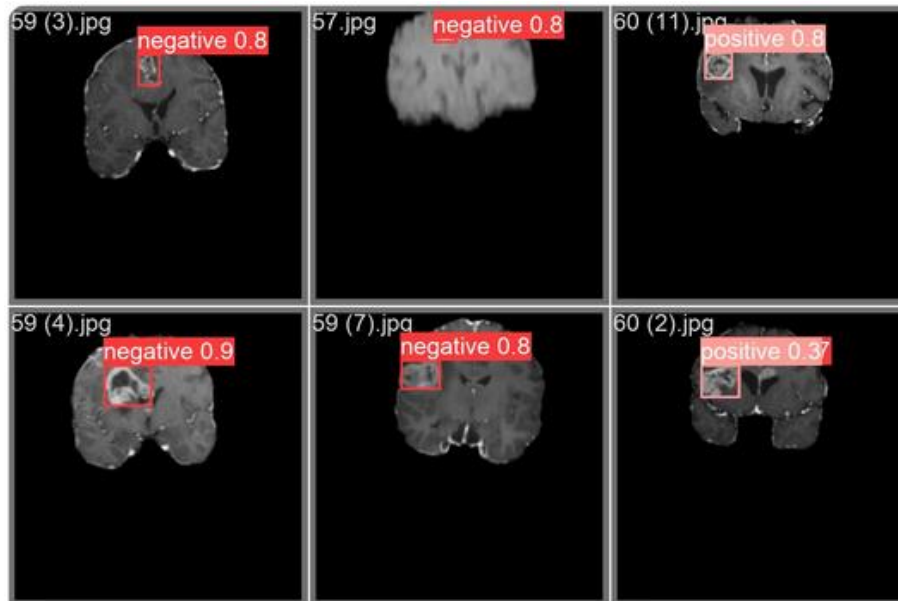
<https://waymo.com/open>



Object Detection: Medical datasets

Brain Tumor Dataset

<https://docs.ultralytics.com/datasets/detect/brain-tumor>



Angiographic dataset for stenosis detection

<https://data.mendeley.com/datasets/ydrm75xywg/3>
<https://data.mendeley.com/datasets/ydrm75xywg/1>
<https://www.nature.com/articles/s41598-021-87174-2/figures/1>

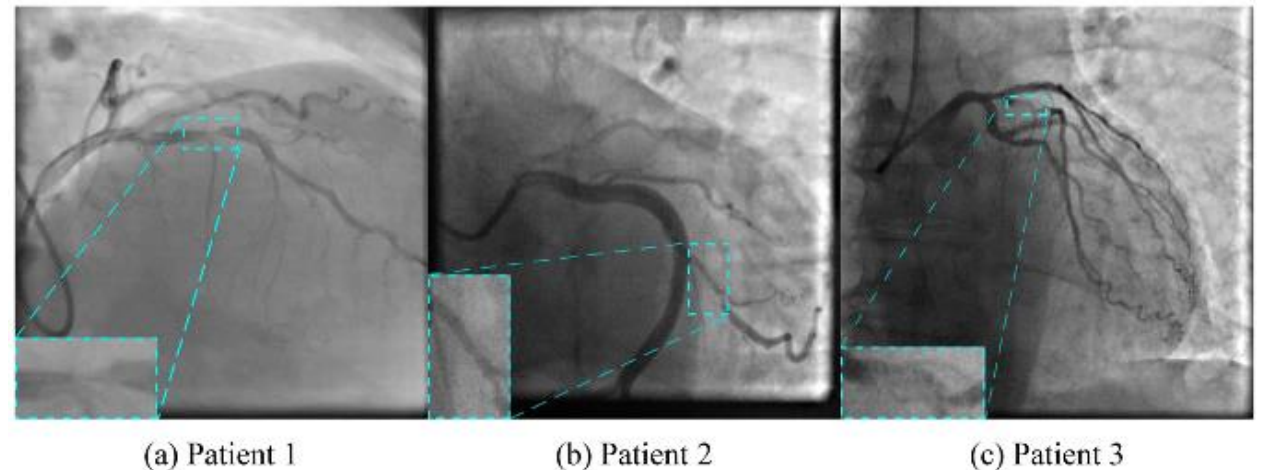


Figure 1. Data labelling of the source images with the callouts of the detected stenotic lesions.

Object Detection: Indoor Objects datasets

HomeObjects-3K Dataset

<https://docs.ultralytics.com/datasets/detect/homeobjects-3k>



Indoor Objects Detection Dataset

<https://datasetninja.com/indoor-object-detection>

<https://www.kaggle.com/datasets/thepbordin/indoor-object-detection/>



Instance Segmentation: Aerial images

LVIS Dataset

<https://docs.ultralytics.com/datasets/detect/lvis>



Carparts Segmentation Dataset

<https://docs.ultralytics.com/datasets/segment/carparts-seg>



Single dataset for Solo Project

Object detection: SKU-110k Dataset

Retail shelf images

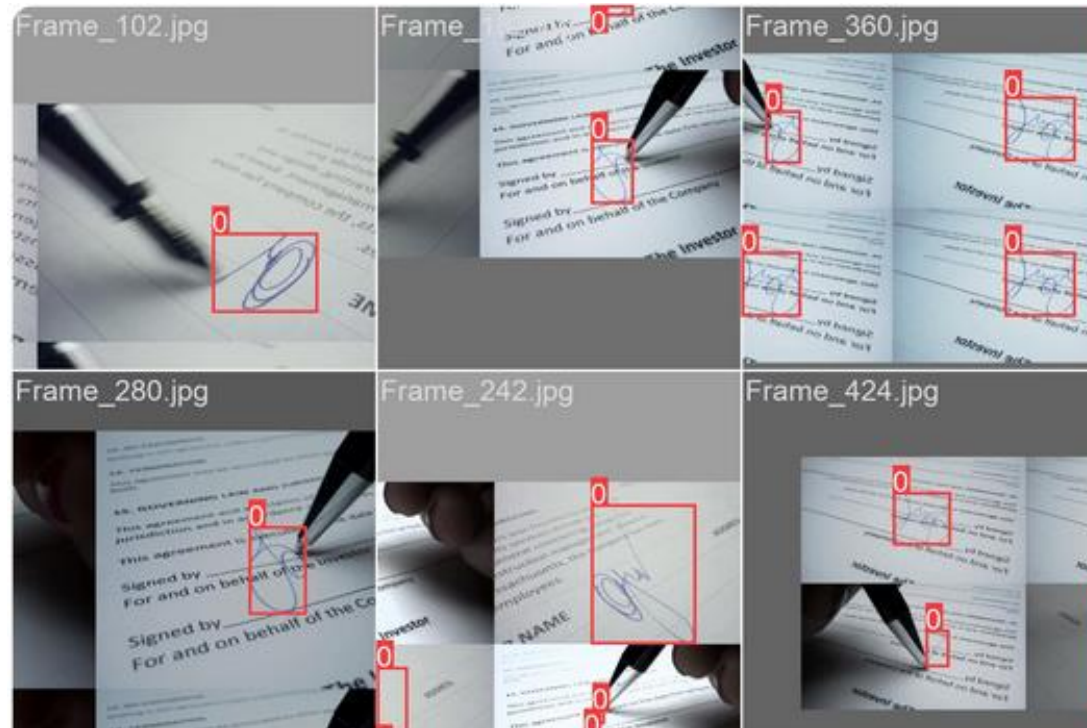
<https://docs.ultralytics.com/datasets/detect/sku-110k>



Object detection: Signature Detection Dataset

Human written signatures within documents

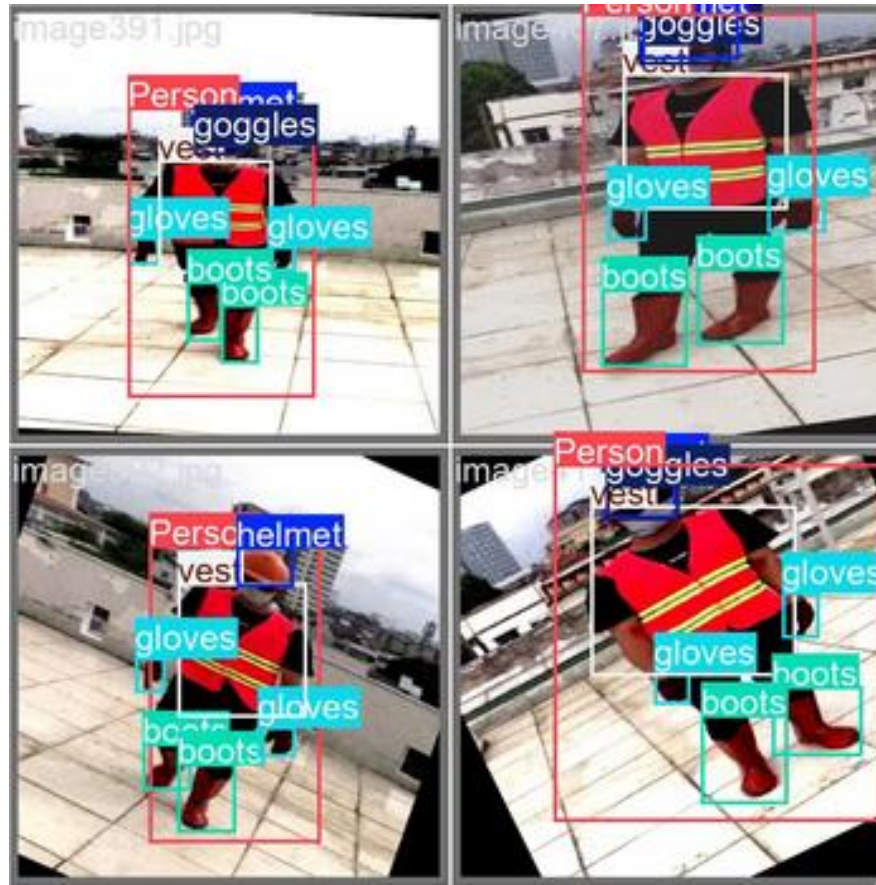
<https://docs.ultralytics.com/datasets/detect/signature>



Object detection: Construction-PPE Dataset

Detect the essential protective gear such as helmets, vests, gloves, boots, and goggles:

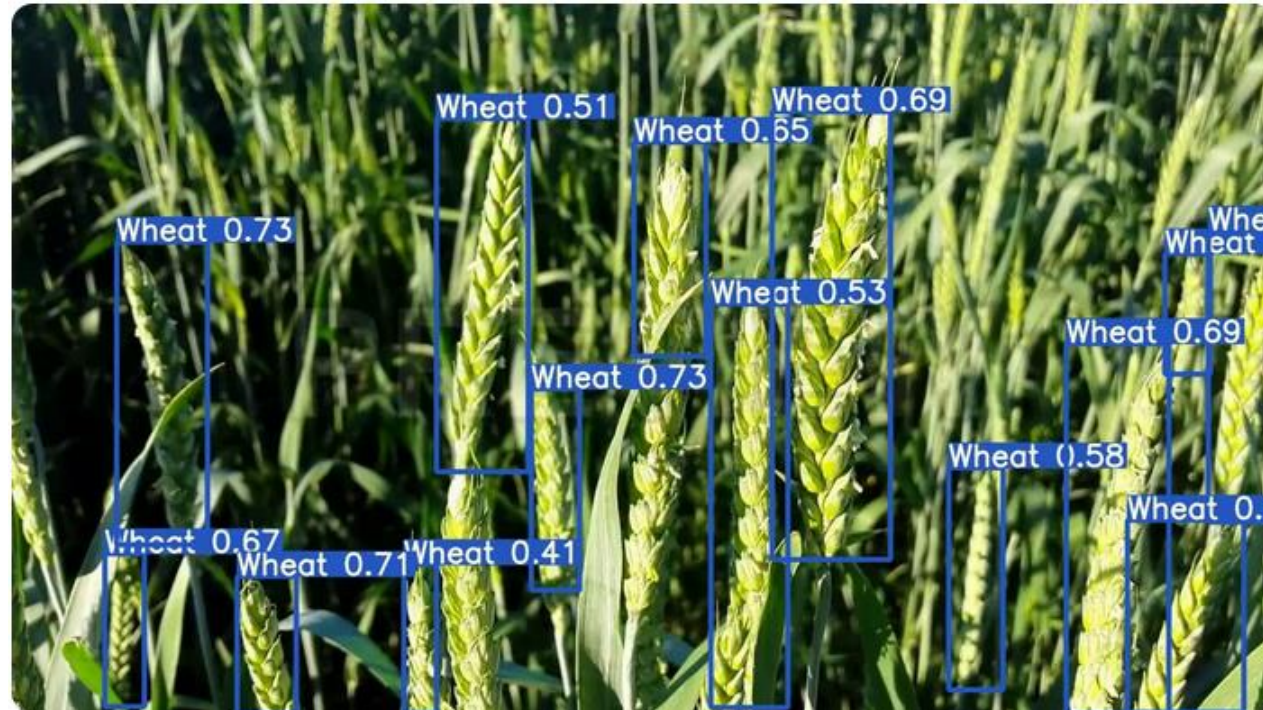
<https://docs.ultralytics.com/datasets/detect/construction-ppe>



Object detection: Global Wheat Head Dataset

Wheat head detection:

<https://docs.ultralytics.com/datasets/detect/globalwheat2020>



Evaluation metrics

- Average Precision: AP [0.5: 0.95],
- Average precision small: APS
- Average precision medium: APM
- Average precision large: APL
- Accuracy Top 1%
- Latency (ms)
- FLOPs (million)
- # of model parameters (million)

During the next few days

- Define team members and select one of datasets categories, define 4 algorithms
- Next lecture: present 2 pages that discuss the importance of object detection for the selected category and put a survey for it
- Define training and testing set according to mechanism used for same dataset in relevant papers.